

Isometric Projection



Objectives

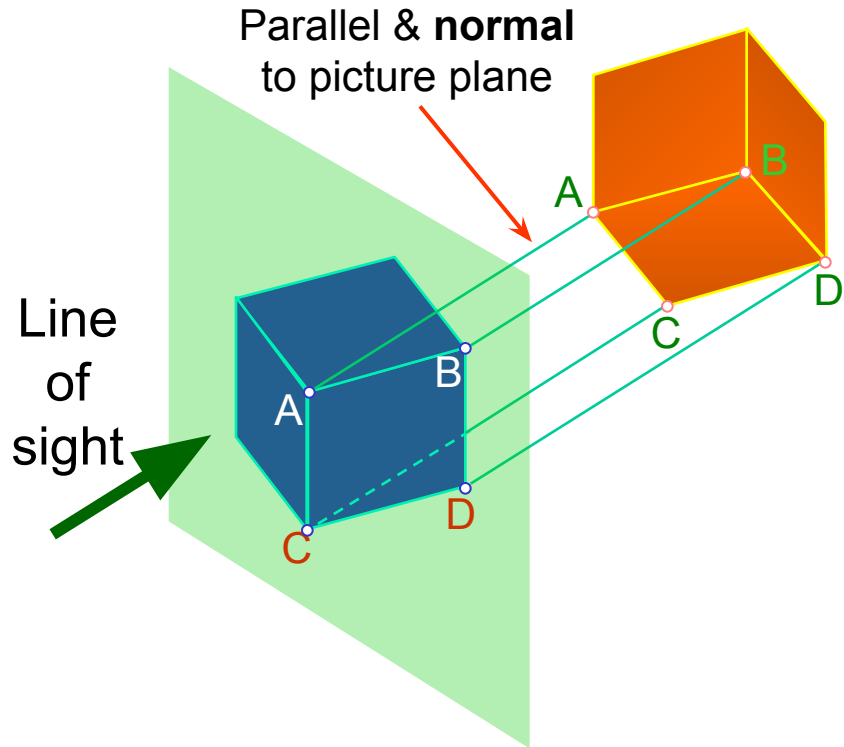
- Be able to **explain the difference** between an *axonometric projection* and an *oblique projection*.
- Be able to **explain the difference** between an *isometric projection* and an *isometric drawing/sketch*.
- Be able to **create** an *isometric* drawing from an actual object and multiview drawing.



Axonometric & Oblique Projection

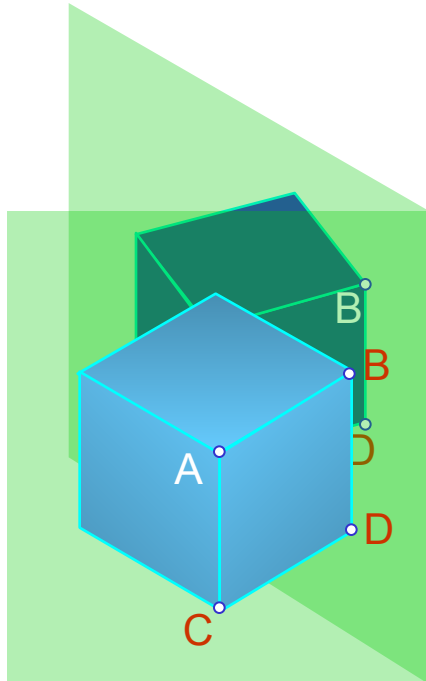


Axonometric Projection

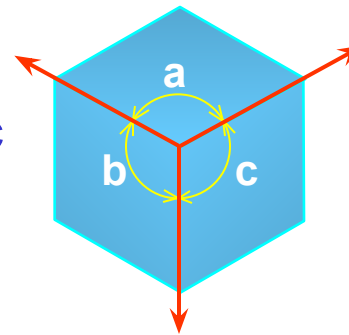


Axonometric Projection

Type of axonometric drawing



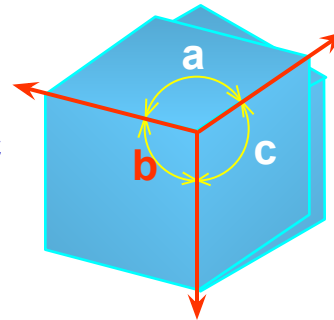
1. Isometric



Axonometric axis

All angles are equal.

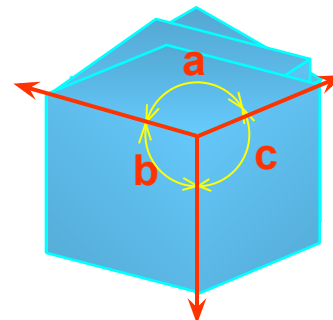
2. Dimetric



Axonometric axis

Two angles are equal.

3. Trimetric

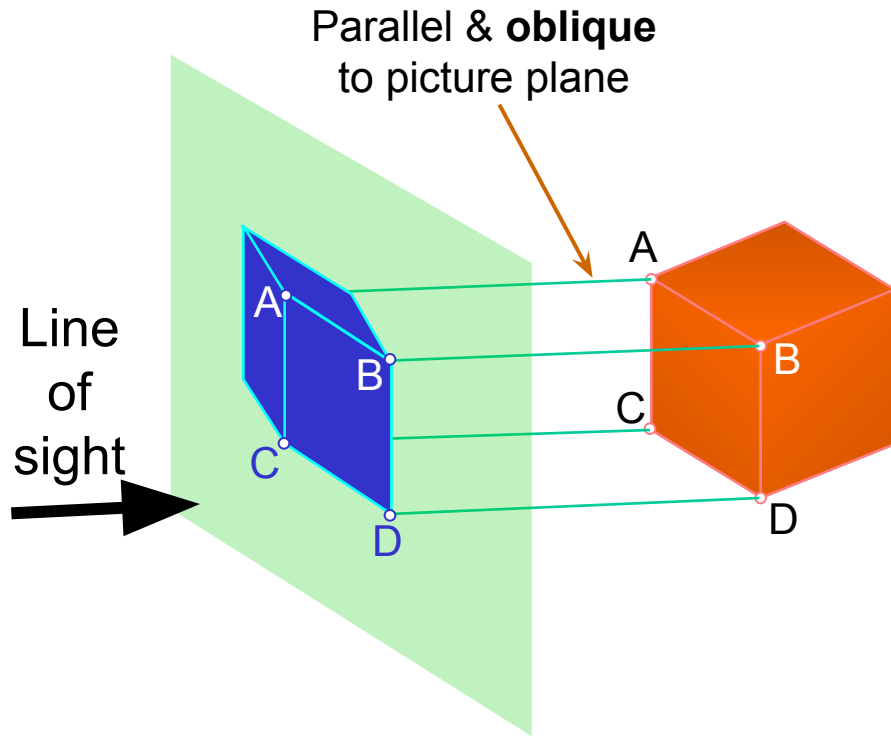


Axonometric axis

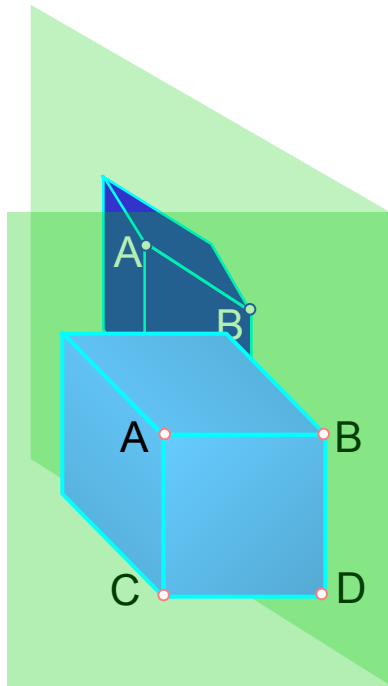
None of angles are equal.



Oblique Projection



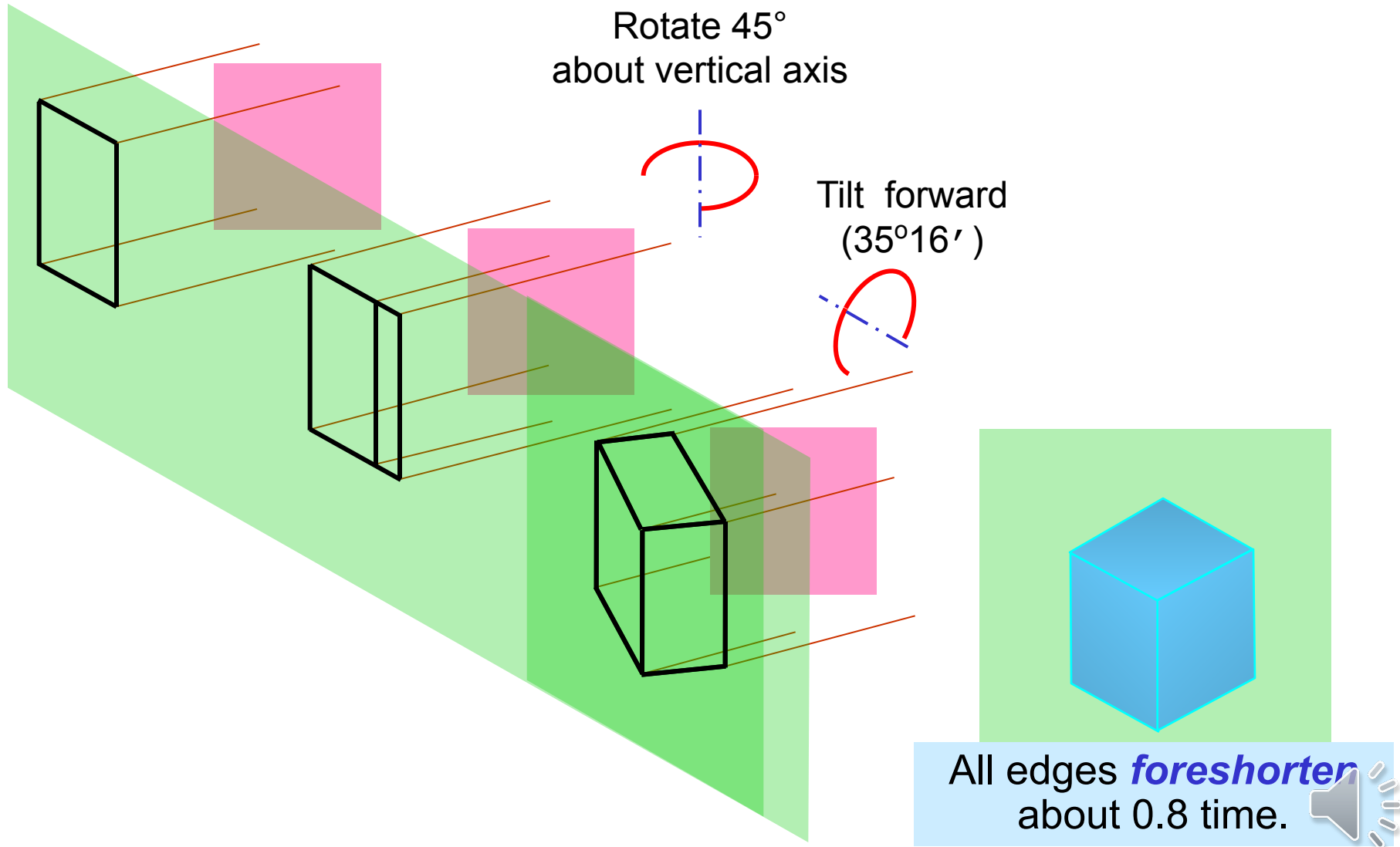
Oblique Projection



Isometric Projection & Isometric drawing



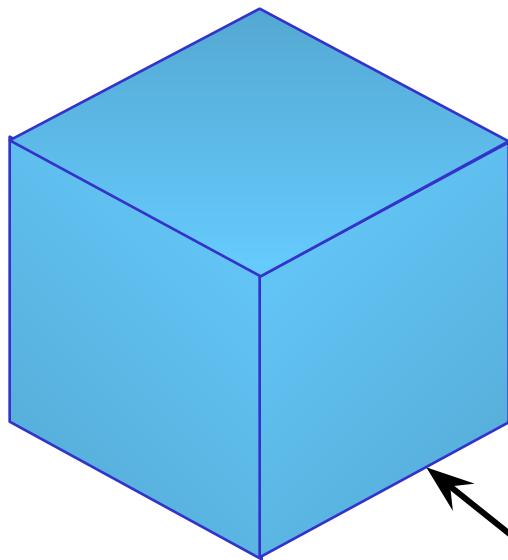
Isometric Projection



Isometric Drawing

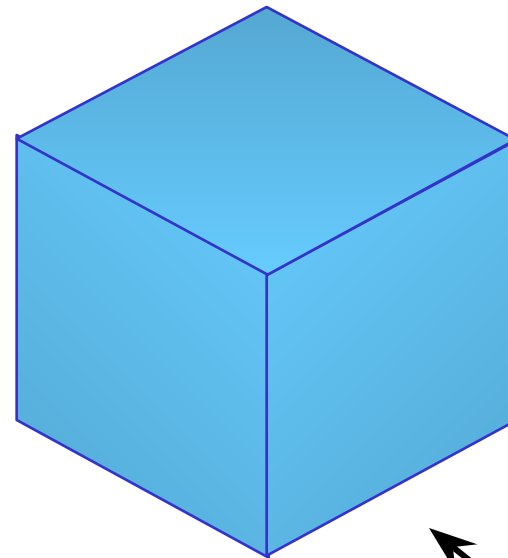
Isometric drawing is a drawing drawn on an isometric axes using *full scale*.

Isometric projection
(True projection)



Forshorten

Isometric drawing
(Full scale)



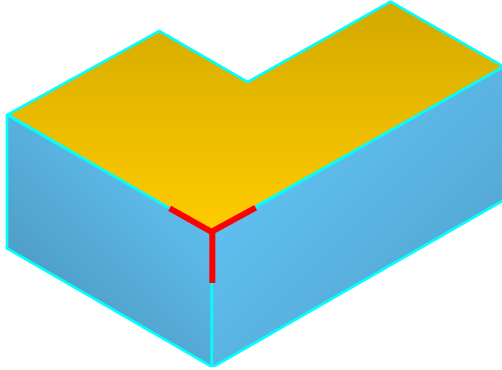
Full scale



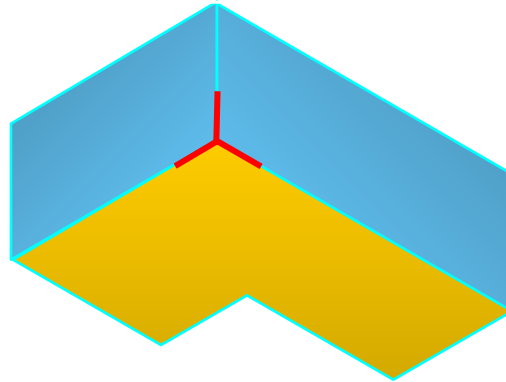
Positions of Isometric Axes

Isometric axes can be arbitrarily positioned to create different views of a single object.

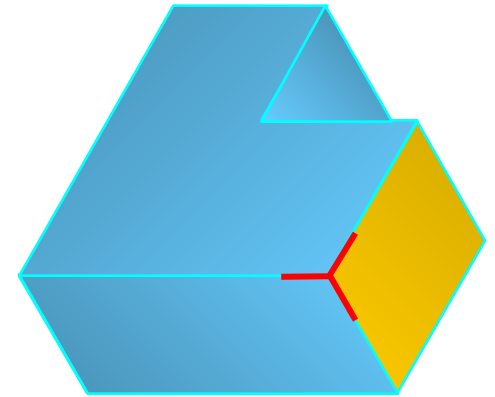
**Regular
isometric**



**Reverse axis
isometric**



**Long axis
isometric**



View point is looking down on the top of the object.

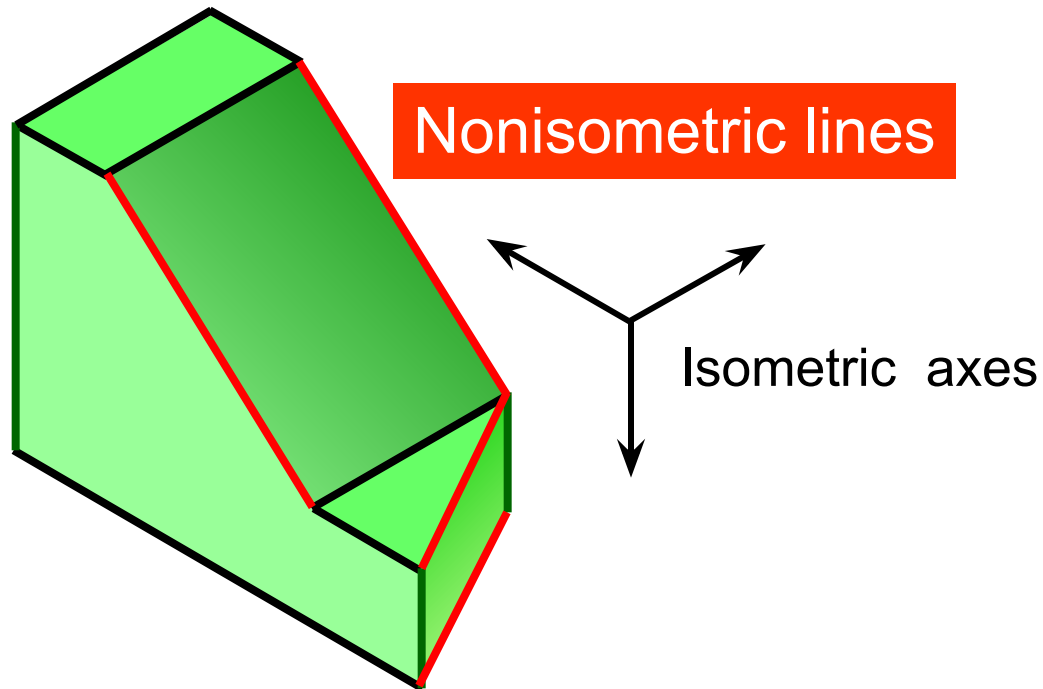
View point is looking up on the bottom of the object.

View point is looking from the right (or left) of the object.



Distance in Isometric Drawing

- **True-length distances** are shown along isometric lines.
- ***Isometric line*** is the line that run ***parallel*** to any of the isometric axes.



Isometric Sketching



Sketch from an actual object

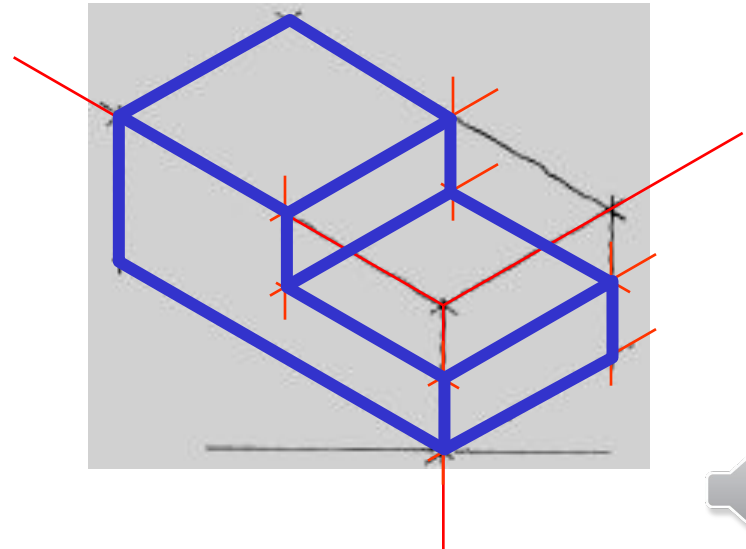
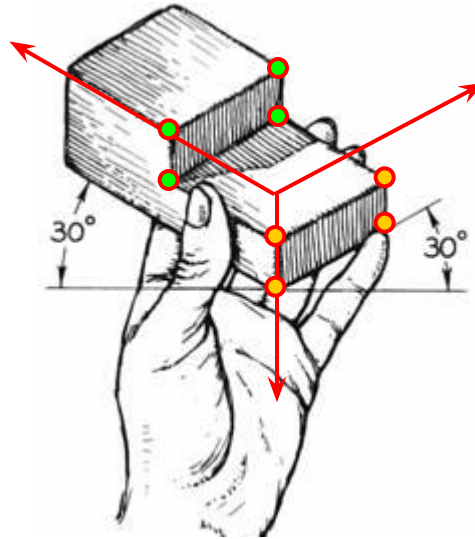
1. Place the object in the position which its shape and features are clearly seen.
2. Define an isometric axis.
3. Sketching the enclosing box.
4. Estimate the size and relationship of each details.
5. Darken all visible lines.



Sketch from an actual object

STEPS

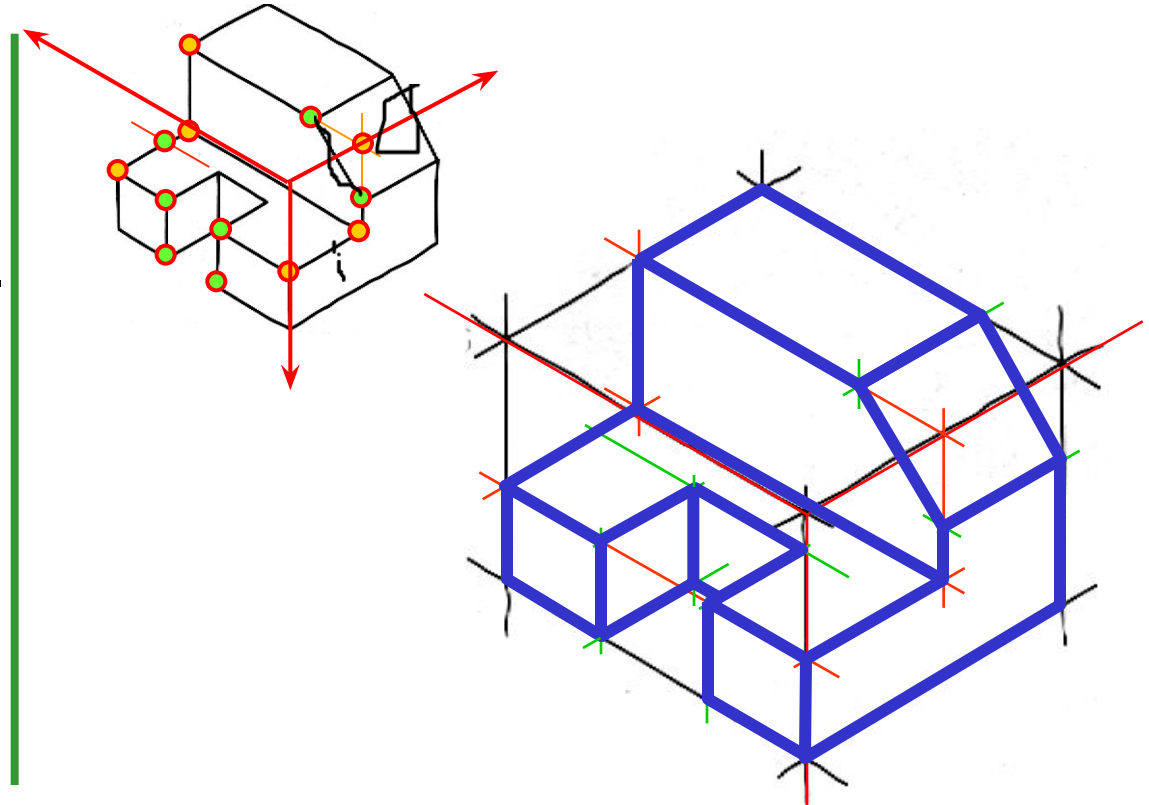
1. Positioning object.
2. Select isometric axis.
3. Sketch enclosing box.
4. Add details.
5. Darken visible lines.



Sketch from an actual object

STEPS

1. Positioning object.
2. Select isometric axis.
3. Sketch enclosing box.
4. Add details.
5. Darken visible lines.



Note In isometric sketch/drawing), hidden lines are *omitted* unless they are absolutely necessary to completely describe the object.

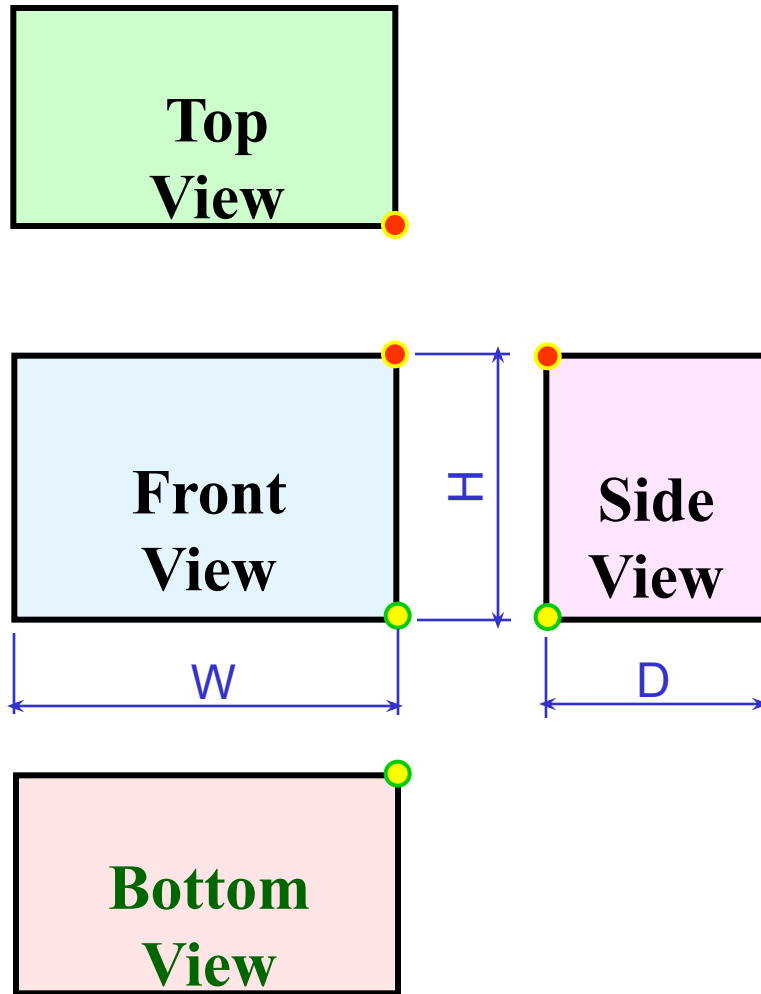


Sketch from multiview drawing

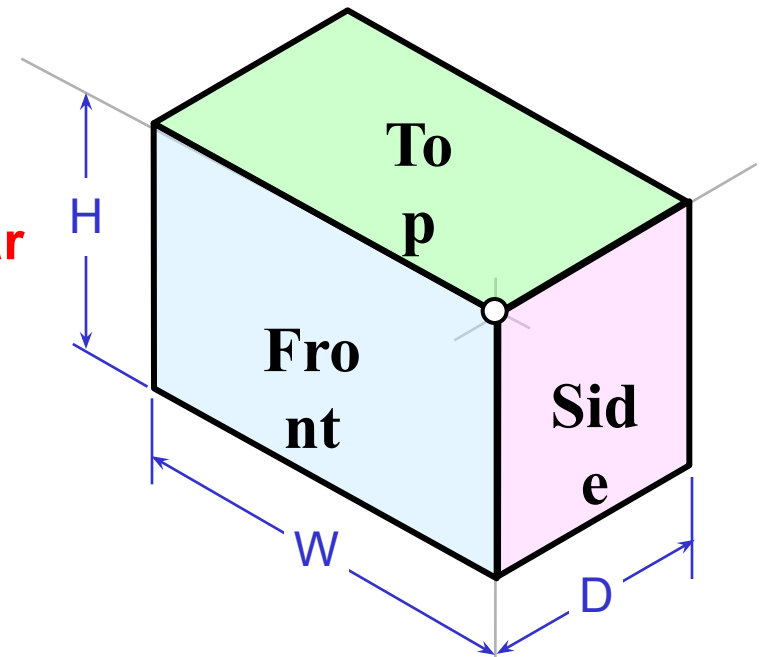
1. Interpret the *meaning of lines/areas* in multiview drawing.
2. Locate the lines or surfaces relative to isometric axis.



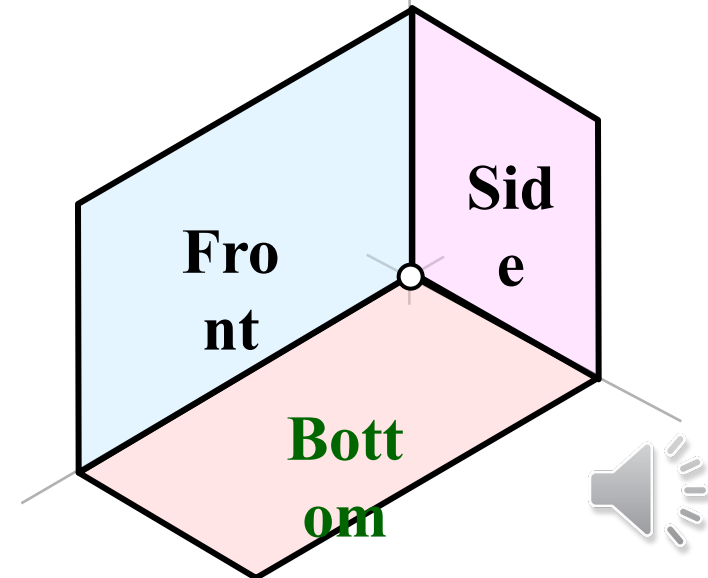
Example 1 : Object has only normal surfaces



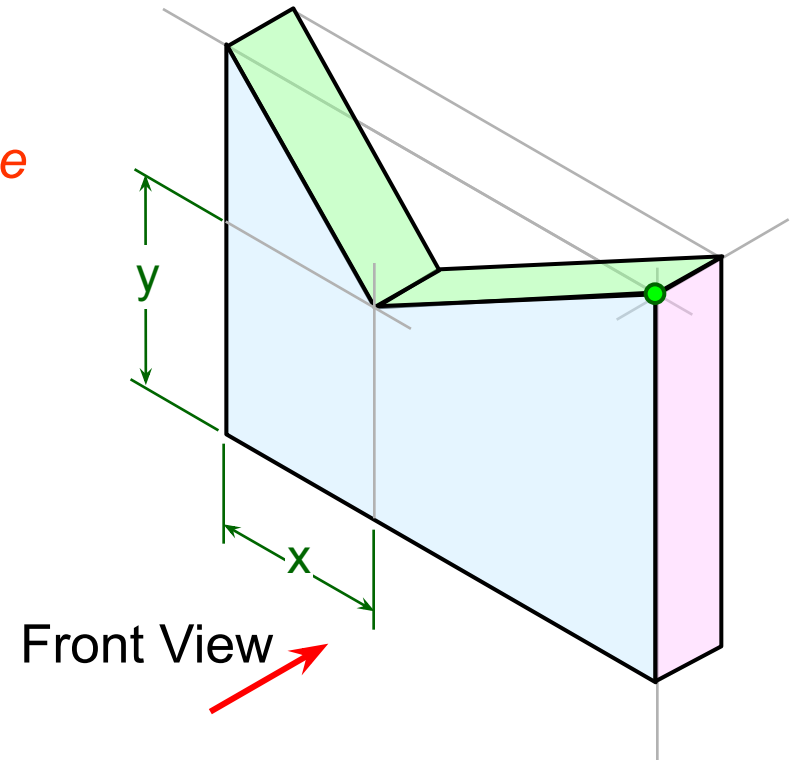
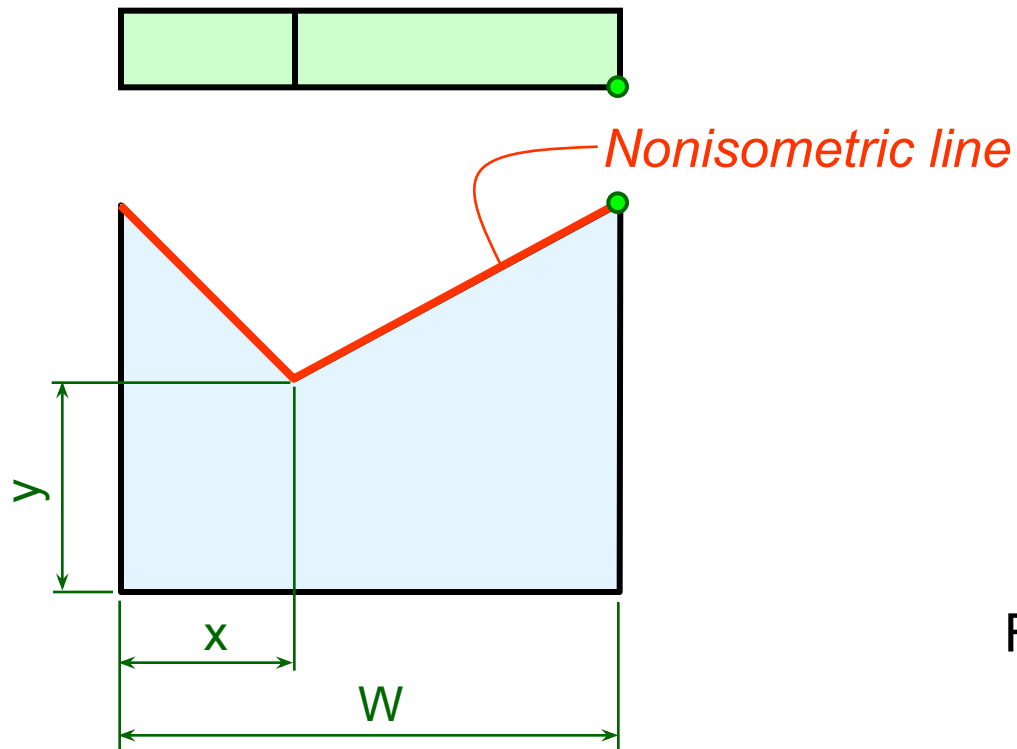
Regular



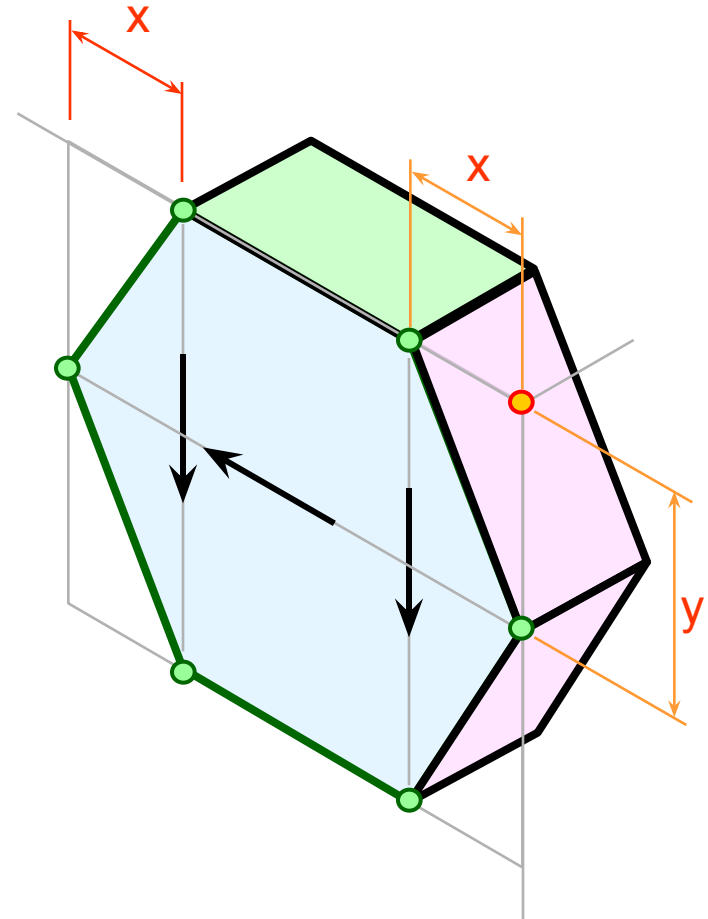
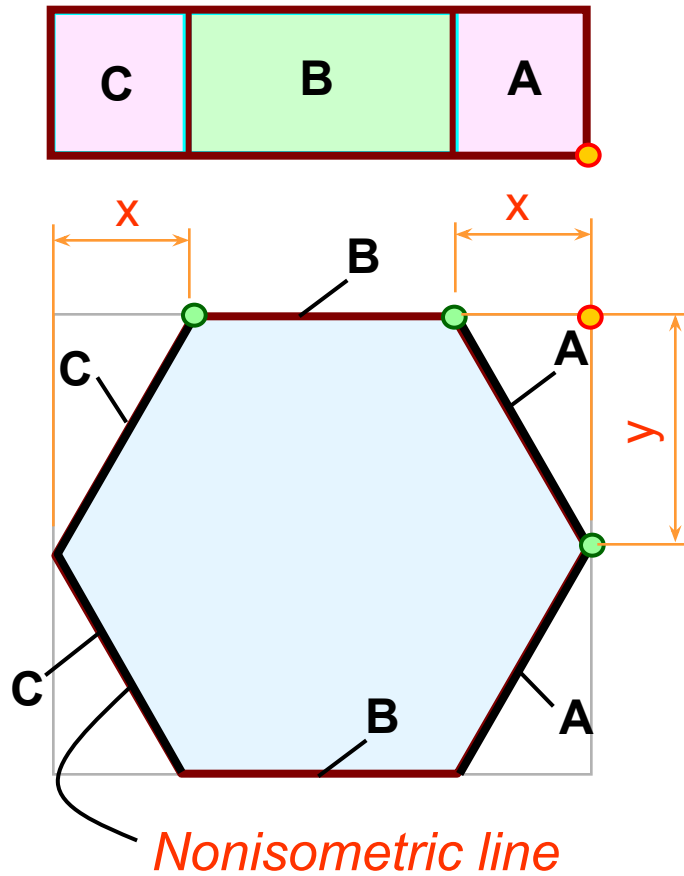
Reverse



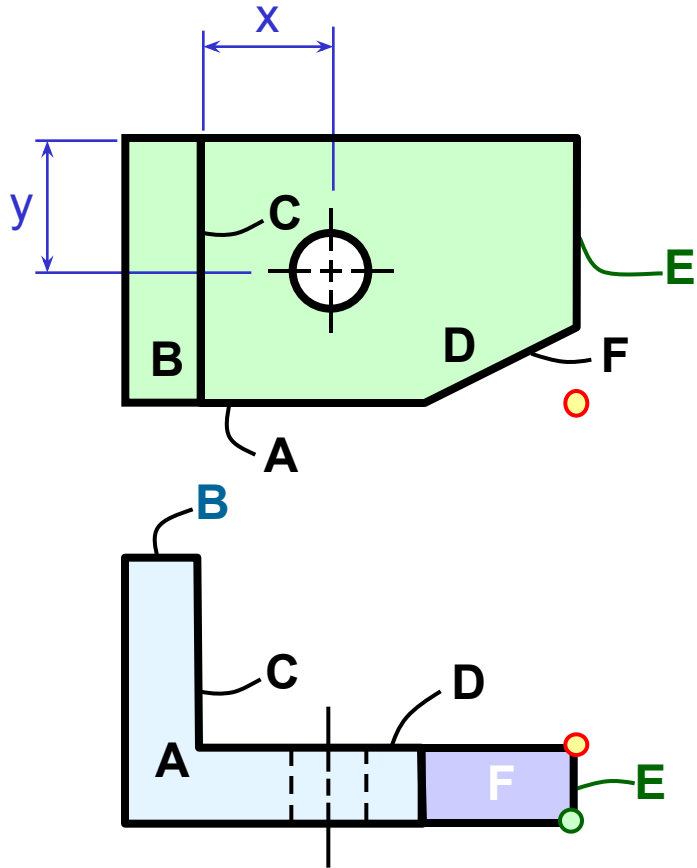
Example 2 : Object has inclined surfaces



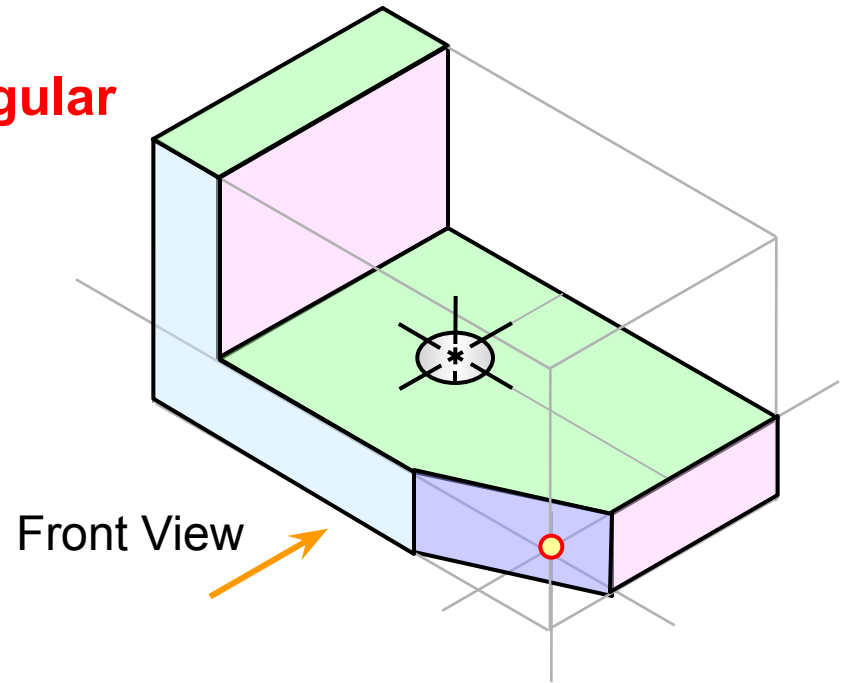
Example 3 : Object has inclined surfaces



Example 4

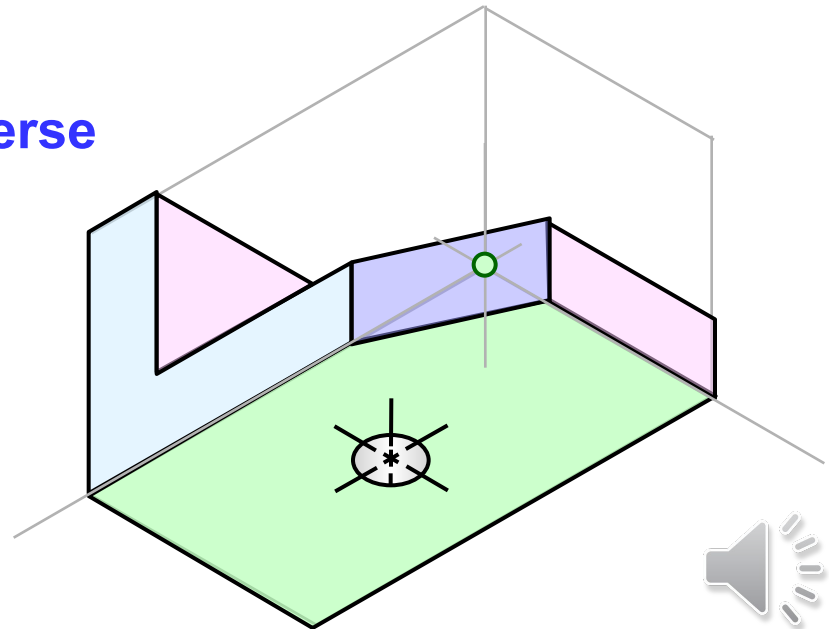


Regular



Front View

Reverse

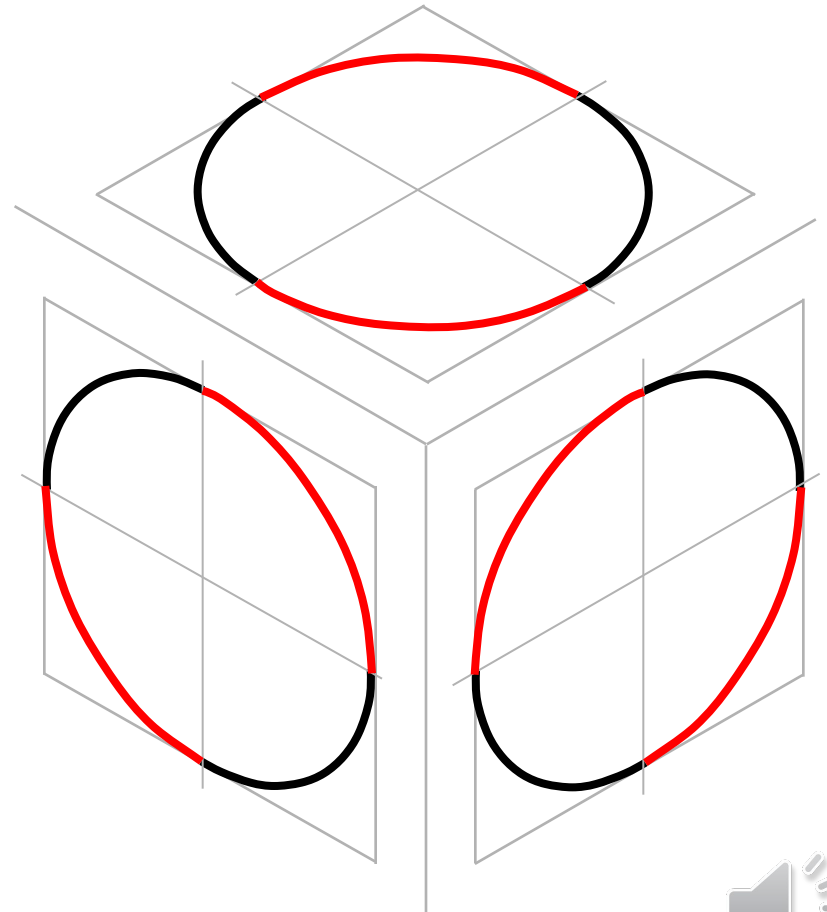


Circle in Isometric

■ In isometric drawing, a circle appears as an ellipse.

Sketching Steps

1. Locate the center of an ellipse.
2. Construct an isometric square.
3. Sketch arcs that connect the tangent points.



Example 5

